

SOMERSET, NJ, USA

WMO: 722247

Lat: 40.624N

Lon: 74.669W

Elev: 105

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54785

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.8	(c) 12.4	(d) -6.5	(e) 3.9	(f) 12.6	(g) -1.8	(h) 5.0	(i) 17.3	(j) 20.6	(k) 28.7	(l) 18.7	(m) 28.9	(n) 3.1	(o) 310	(p) 0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	91.6	74.6	89.3	73.4	86.8	71.9	77.4	87.1	76.0	84.5	74.7	82.4	5.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	130.9	81.5	73.3	124.2	79.6	72.3	120.0	78.7	40.9	87.2	39.4	84.5	38.1	82.6	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.3	(b) 15.0	(c) 12.3	(e) 1.8	(f) 95.9	(g) 6.7	(h) 3.3	(i) -3.0	(j) 98.3	(k) -7.0	(l) 100.2	(m) -10.8	(n) 102.1	(o) -15.6	(p) 104.5		
(4) 17.3	(5) 15.0	(6) 12.3	(7) 1.8	(8) 95.9	(9) 6.7	(10) 3.3	(11) -3.0	(12) 98.3	(13) -7.0	(14) 100.2	(15) -10.8	(16) 102.1	(17) -15.6	(18) 104.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	29.8	32.0	40.2	51.6	61.2	69.7	74.8	73.2	66.6	54.5	44.4	35.6	(6)
(7)		DBStd	17.42	9.76	8.97	8.80	8.36	7.50	5.97	4.73	4.77	6.36	8.34	8.28	8.62	(7)
(8)		HDD50	2247	626	507	322	76	5	0	0	0	48	209	454		(8)
(9)		HDD65	5314	1090	925	768	409	167	25	1	2	56	341	619	911	(9)
(10)		CDD50	3313	2	3	19	125	353	590	769	719	497	187	41	8	(10)
(11)		CDD65	903	0	0	0	8	51	165	305	257	103	14	0	0	(11)
(12)		CDH74	8695	0	1	9	189	615	1652	2942	2259	916	109	3	0	(12)
(13)		CDH80	3105	0	0	0	53	188	565	1197	806	277	19	0	0	(13)
(14)	Wind	WSAvg	4.0	4.7	5.0	5.7	5.3	3.7	3.1	2.8	2.5	3.1	3.6	4.3	4.4	(14)
(15)	Precipitation	PrecAvg	50.3	3.6	2.9	4.5	3.9	3.9	4.8	4.9	4.5	4.7	4.6	3.5	4.6	(15)
(16)		PrecMax	65.2	7.7	6.5	10.0	10.8	9.0	9.6	9.7	9.2	12.3	13.6	6.6	8.3	(16)
(17)		PrecMin	40.1	1.7	0.9	0.9	1.5	0.8	1.1	1.1	1.4	1.1	0.7	1.0	1.1	(17)
(18)		PrecStd	6.1	1.5	1.3	1.9	2.0	1.7	2.2	2.2	2.0	2.6	3.1	1.5	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	65.8	74.3	86.5	89.3	92.8	95.8	94.0	91.7	82.3	72.2	66.2	(19)
(20)			MCWB	56.1	55.5	59.3	64.8	71.3	74.6	75.4	76.1	73.7	68.9	61.8	60.2	(20)
(21)		2%	DB	55.1	57.2	66.3	79.3	85.3	89.3	92.3	90.3	86.9	76.8	67.6	59.0	(21)
(22)			MCWB	50.4	48.2	54.4	60.8	69.4	73.4	74.9	74.6	70.7	65.8	59.8	54.8	(22)
(23)		5%	DB	48.3	50.7	60.5	73.4	81.2	86.0	89.6	87.4	82.6	72.7	63.4	53.6	(23)
(24)			MCWB	43.0	43.3	49.8	58.5	66.4	71.3	74.0	72.7	68.8	64.1	56.1	49.5	(24)
(25)		10%	DB	44.1	45.2	54.9	68.1	76.5	83.1	86.7	84.5	79.4	69.1	59.2	49.2	(25)
(26)			MCWB	39.5	38.5	46.8	55.2	64.1	69.7	72.6	71.1	67.8	61.5	52.9	44.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	59.2	61.2	68.1	74.0	77.1	79.6	79.8	76.8	72.4	65.2	61.8	(27)
(28)			MCD B	61.2	65.0	71.6	79.9	84.9	88.3	91.1	89.8	86.7	77.3	69.8	64.9	(28)
(29)		2%	WB	50.3	50.4	56.5	64.5	71.5	75.5	77.8	77.3	74.5	69.4	61.6	55.4	(29)
(30)			MCD B	54.3	57.2	64.1	75.7	81.6	85.3	87.8	85.7	81.1	74.7	66.1	58.6	(30)
(31)		5%	WB	44.2	43.7	51.9	60.8	69.1	73.9	76.3	75.5	72.6	66.2	57.9	50.6	(31)
(32)			MCD B	47.4	48.7	58.3	70.6	77.3	82.8	84.9	83.0	78.3	71.1	62.2	53.4	(32)
(33)		10%	WB	39.9	39.7	47.5	57.3	66.6	71.9	74.8	73.9	70.7	62.9	54.1	45.2	(33)
(34)			MCD B	43.8	44.1	54.2	65.5	73.9	79.9	82.7	80.4	76.4	67.0	57.8	48.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.4	19.2	20.0	23.3	22.5	21.9	21.6	21.1	22.2	21.7	20.6	16.6	(35)
(36)			MCD BR	23.8	28.3	29.6	32.9	28.9	25.7	24.9	24.1	26.6	25.9	25.7	22.1	(36)
(37)			MCWBR	19.3	20.3	19.1	16.8	14.5	12.0	10.0	10.1	12.6	15.3	17.6	18.7	(37)
(38)		5% WB	MCD BR	21.7	26.2	25.6	27.7	23.8	22.7	20.5	19.3	20.1	20.4	20.9	19.8	(38)
(39)			MCWBR	18.8	19.9	19.3	16.4	13.0	11.7	9.6	9.4	11.3	13.8	16.9	18.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.325	0.360	0.384	0.430	0.492	0.489	0.480	0.395	0.362	0.328	0.308	(40)	
(41)		taud	2.452	2.400	2.373	2.339	2.240	2.105	2.127	2.143	2.352	2.455	2.506	2.515	(41)	
(42)		Ebn at Noon	270	282	282	282	270	252	252	250	267	265	261	262	(42)	
(43)		Edh at Noon	25	30	35	38	43	50	49	46	35	29	24	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	590	878	1163	1504	1665	1826	1873	1603	1348	943	665	502	(44)	
(45)		RadStd	51	92	95	146	182	161	92	104	150	99	60	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+122

Nomenclature: See separate page